

Lab-9: Network Address Translation (NAT) – Connecting to an ISP

Objectives

- To learn about Network Address Translation (NAT): why and how used?
- Types of NAT: Static and Dynamic NAT, Port Address Translation (PAT)
- To build an internetwork using NAT using Packet Tracer

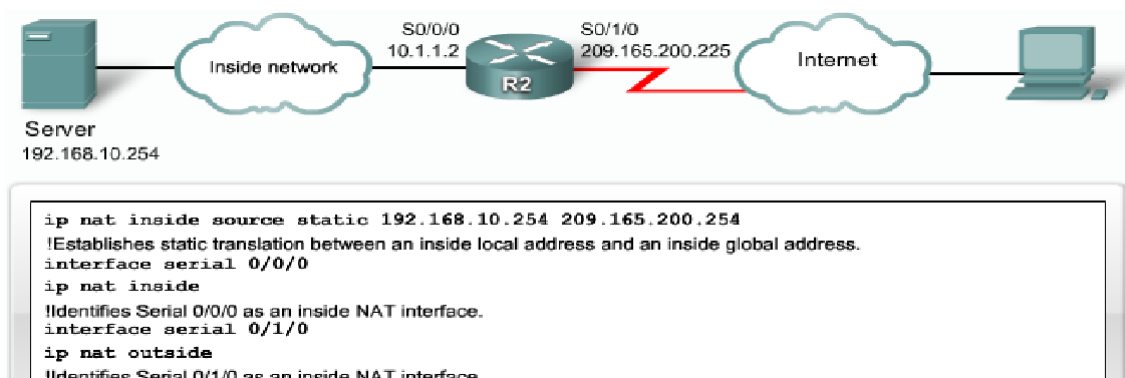
Part-1: Background: Network Address Translation (NAT)

NAT (Network Address Translation) is a technique for preserving scarce Internet IP addresses. It converts private IP addresses (not routable to Internet) to public IP addresses so that Internet can be accessed from private network.

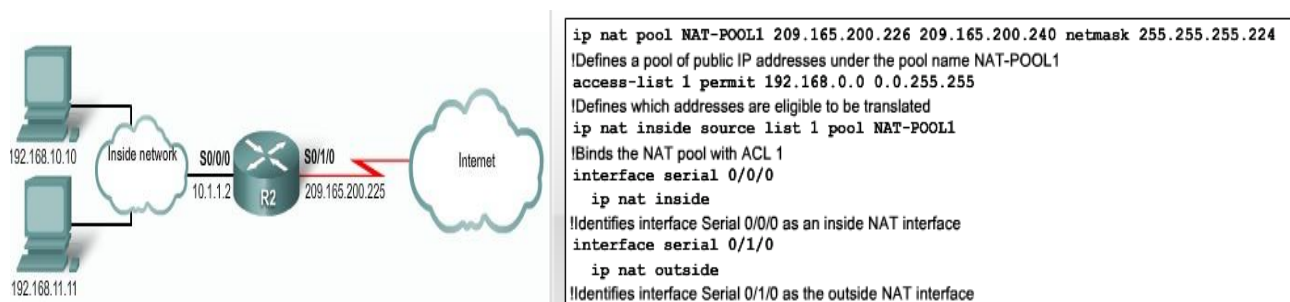
Types of NAT

Developed by Cisco, Network Address Translation is used by a device (firewall, router or computer) that sits between an internal network and the rest of the world. NAT has many forms and can work in several ways:

Static NAT - Mapping an unregistered (private) IP address to a registered (public) IP address on a one-to-one basis. Particularly useful when a local host with private IP address needs to be accessible from outside the network (from Internet).



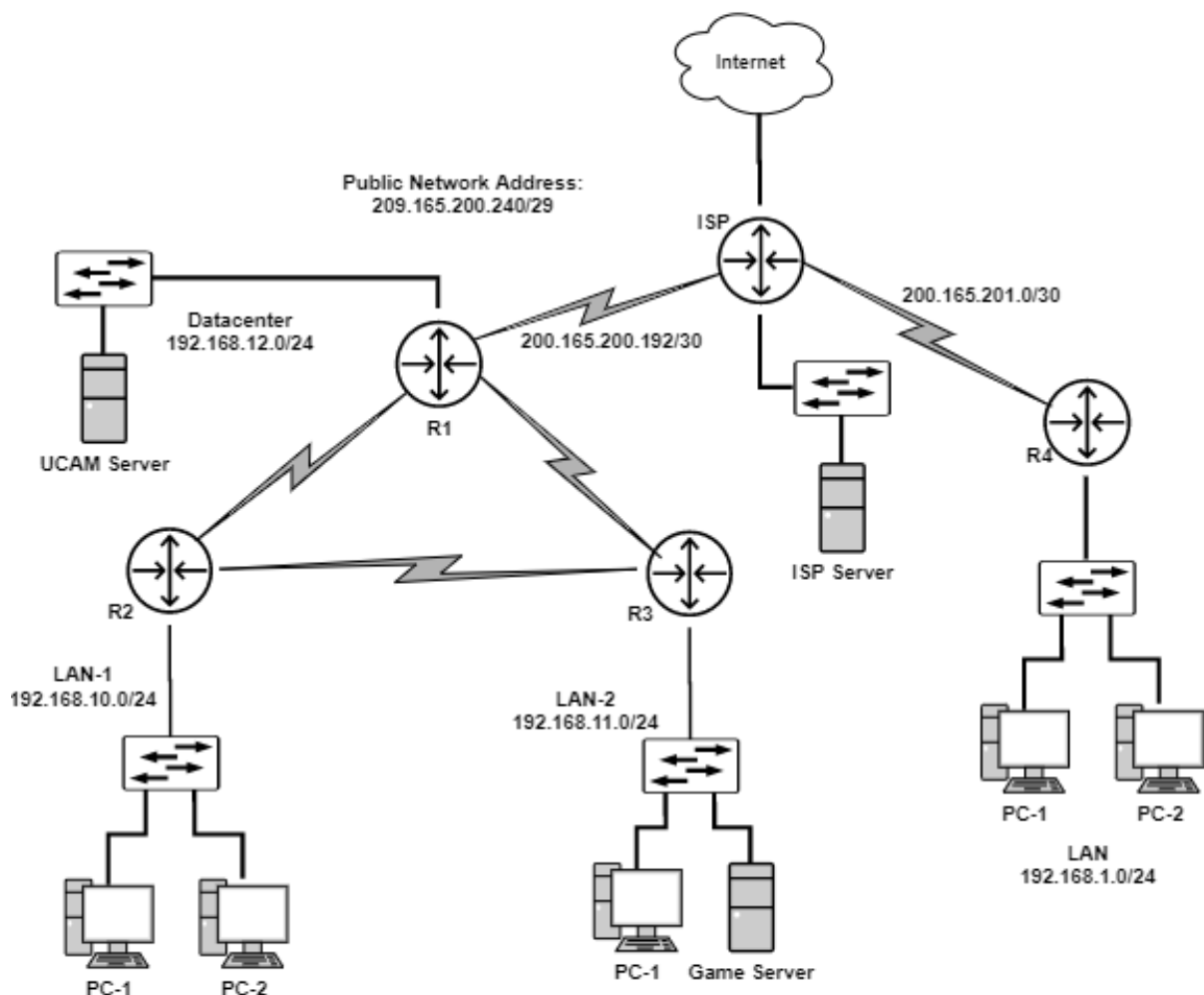
Dynamic NAT - Maps an unregistered IP address to a registered IP address from a group of registered IP addresses.



PAT (Port Address Translation)- A form of dynamic NAT that maps multiple unregistered IP addresses to a single registered IP address by using different ports. This is known also as PAT (Port Address Translation), single address NAT or port-level multiplexed NAT.

```
access-list 1 permit 192.168.0.0 0.0.255.255
!Defines which addresses are eligible to be translated
ip nat inside source list 1 interface serial 0/1/0 overload
!Identifies the outside interface Serial 0/1/0 as the inside global address to be overloaded
interface serial 0/0/0
  ip nat inside
!Identifies Serial 0/0/0 as an inside NAT interface.
interface serial 0/1/0
  ip nat outside
!Identifies Serial 0/1/0 as an inside NAT interface.
```

Part-2: Configure Network Address Translation (NAT)



Part-2(A): Configure Routing

ISP (Internet Service Provider) uses static routing to reach all networks beyond R1 (Border routers). However, R1 translate private addresses into public addresses before sending traffic to ISP (Internet). Therefore, ISP must be configured with the public addresses (209.165.200.240/29) for R1. But R4 no need to ISP routing because it's use the link public address for PAT.

We will show the configurations on R1, Enter the following static route on ISP. This static route includes all addresses assigned to R1 for public use.

```
ISP (config) #ip route 209.165.200.240 255.255.255.248 200.165.200.194
```

Configure RIP in all LAN routers,

```
R1(config)#router rip
R1(config-router)#net 192.168.12.0
R1(config-router)#net 192.168.220.0
R1(config-router)#net 192.168.221.0
R1(config-router)#exit
```

Note: Configure others LAN router.

Configure a default route on R1 and propagate the route in RIP.

```
R1(config)#ip route 0.0.0.0 0.0.0.0 209.165.200.193

R1(config)#router rip
R1(config-router)#default-information originate
R1(config-router)#exit
```

Allow a few seconds for R2 & R3 to learn the default route from R1 and then check the R2 & R3 routing table using show ip route command.

Alternatively, you can clear the routing table with the clear ip route "*" command. A default route pointing to R1 should appear in the R2 & R3 routing table. From R2, ping the serial interface (connected to ISP) on R1. The pings should be successful. Troubleshoot if the pings fail.

Part-2(B): Configure Static NAT

Statically map a public IP address to a private IP address.

The **UCAM Server** attached to R1 is accessible by outside hosts beyond ISP. Statically assign the public IP address 209.165.200.241 as the address for NAT to use to map packets to the private IP address of the **UCAM Server** at 192.168.12.12. Similarly, assign the public IP address 209.165.200.242 to the private IP address of the **Game Server** at 192.168.11.11

```
R1 (config)#ip nat inside source static 192.168.12.12 209.165.200.241
R1 (config)#ip nat inside source static 192.168.11.11 209.165.200.242
```

Part-2(C): Configure Dynamic NAT

While static NAT provides a permanent mapping between an internal address and a specific public address, dynamic NAT maps private IP addresses to public addresses. These public IP addresses come from a NAT pool.

Define a pool of global addresses.

Create a pool of addresses to which matched source addresses are translated. The following command creates a pool named **NAT-POOL** that translates matched addresses to an available IP address in the 209.165.200.243 - 209.165.200.245 range.

```
R1 (config)#ip nat pool NAT-POOL 209.165.200.243 209.165.200.245 netmask 255.255.255.248
```

Create a standard access control list to identify which inside addresses are translated.

```
R1 (config)#access-list 1 permit 192.168.10.0 0.0.0.255
R1 (config)#access-list 1 permit 192.168.11.0 0.0.0.255
R1 (config)#access-list 1 permit 192.168.12.0 0.0.0.255
```

Establish dynamic source translation by binding the pool with the access control list.

A router can have more than one NAT pool and more than one ACL. The following command tells the router which address pool to use to translate hosts that are allowed by the ACL.

```
R1 (config)#ip nat inside source list 1 NAT pool NAT-POOL
```

Part-2(D): Specify inside and outside NAT interfaces

Before NAT can work, you must specify which interfaces are inside and which interfaces are outside.

```
R1(config)#interface serial 0/1/1
R1(config-if)#ip nat outside
R1(config-if)#exit

R1(config-if)#interface G0/0
R1(config-if)#ip nat inside
R1(config-if)#exit

R1(config)#interface serial 0/0/0
R1(config-if)#ip nat inside
R1(config-if)#exit

R1(config)#interface serial 0/0/1
R1(config-if)#ip nat inside
R1(config-if)#exit
```

Part-2(E): Verify the configuration.

```
R1#show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
---	209.165.200.241	192.168.12.2	---	---
---	209.165.200.242	192.168.11.3	---	---
icmp	209.165.200.243	192.168.10.2	200.165.200.193	200.165.200.193
icmp	209.165.200.244	192.168.11.2	200.165.200.193	200.165.200.193

Part-3: Configure PAT (Port Address translation)

In the previous example, what would happen if you needed more than the four public IP addresses that the pool allows? By tracking port numbers, NAT overloading allows multiple inside users to reuse a public IP address.

In this task, you will configure **PAT** (or NAT overload) on R4 so that all internal IP addresses are translated to the R4 S0/0/0 IP address when connecting to any outside device.

The **configuration is similar to dynamic NAT**, except that instead of a pool of addresses, the interface keyword is used to identify the outside IP address. Therefore, no NAT pool is defined. The overload keyword enables the addition of the port number to the translation.

Create a standard access control list to identify which inside addresses are translated.

```
R4 (config)#access-list 10 permit 192.168.1.0 0.0.0.255
```

Establish dynamic source translation by binding the pool with the access control list.

```
R4(config)#ip nat inside source list 10 interface S0/0/0 overload
```

Specify inside and outside NAT interfaces

```
R4(config)#int serial 0/0/0
R4(config-if)#ip nat outside
R4(config-if)#exit

R4(config)#int gig0/0
R4(config-if)#ip nat inside
R4(config-if)#exit
```

Verify the configuration

```
R4#show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	209.165.200.194:3	192.168.10.11:3	209.165.200.193:3	209.165.200.193:3
icmp	209.165.200.194:1024	192.168.11.11:3	209.165.200.193:3	209.165.200.193:1024
---	209.165.200.241	192.168.12.2 ---	---	
---	209.165.200.242	192.168.11.3 ---	---	